

devine-1989-stereotypes-and-prejudice--their-automatic-a

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Paper Synthesis

Paper: devine-1989-stereotypes-and-prejudice--their-automatic-and-controlled-components

Executive Summary

The paper investigates whether automatic activation of Black stereotypes and controlled inhibition of those activations differ between high- and low-prejudice individuals. It finds that both groups possess equivalent knowledge of the stereotype and experience equally strong automatic activation, but only low-prejudice participants can deliberately suppress stereotype-congruent thoughts, supporting a dual-process model that separates automatic prejudice from controlled regulation.

Reading Recommendation

Read Selectively

The paper offers a seminal dual-process test of automatic stereotype activation versus controlled inhibition, making the experimental sections (especially Experiments 2 and 3) worth a detailed read. However, the introduction's claim about personal beliefs is never operationalized and methodological details (power analysis, randomization) are sparse, so the abstract and discussion suffice for a general overview.

Background and Motivation

The paper addresses the gap that existing literature has failed to clearly separate stereotypes from personal beliefs and to examine how each uniquely contributes to intergroup responses. This gap is motivated by classic prejudice frameworks that equated stereotype knowledge with prejudice (e.g., Allport 1954; Billig 1985) and by early cognitive accounts of automatic versus controlled processing (Posner & Snyder 1975; Schneider & Shiffrin 1977).

Theoretical Stance

Overall stance: Consistent Theoretical

The paper introduces the automatic-controlled model in the introduction and repeatedly aligns its empirical findings (equal stereotype knowledge, automatic activation, and controlled inhibition) with that model in the discussion, while consistently opposing the inevitability framework.

Theories Discussed

- **Automatic vs Controlled Processes Model** (*focal*): A dual-process framework derived from information-processing work (Posner & Snyder 1975; Schneider & Shiffrin 1977) that separates involuntary, automatic activation of stereotypes from voluntary, controlled inhibition of those activations.
- **Inevitability of Prejudice Framework** (*alternative*): The view, articulated by Allport (1954), Billig (1985), Ehrlich (1973), Hamilton (1981), and Tajfel (1981), that the mere existence of stereotype knowledge inevitably produces prejudiced responses.
- **Limited-Capacity Processing Perspective** (*background*): The notion, invoked in the discussion, that all individuals are constrained by limited cognitive capacity, which can lead to prejudice when automatic processes dominate but does not make prejudice inevitable.

Study Design

- **Number of experiments:** 3
- **Predictors examined:** prejudice level (high vs. low), automatic stereotype activation, controlled inhibition processes
- **Outcomes examined:** knowledge of the Black cultural stereotype (proportion of listed stereotype components), evaluation of ambiguous stereotype-relevant behavior (stereotype-congruent vs. incongruent judgments), thought-listing performance (frequency of stereotype-congruent vs. equality-oriented thoughts)

- **Materials:** Paper-and-pencil questionnaires: (1) blank sheet for free-response listing of Black stereotype components; (2) ambiguous behavior descriptions without explicit racial cues; (3) thought-listing prompts requiring participants to generate spontaneous thoughts about a target. The Modern Racism Scale was used to classify participants as high or low prejudice.
- **Design types:**
 - exp_1: between-subjects 2 (prejudice level: high vs. low) design assessing stereotype knowledge
 - exp_2: within-subjects design manipulating exposure to a race-unspecified target and measuring automatic activation effects on ambiguous behavior evaluations
 - exp_3: within-subjects thought-listing task assessing controlled inhibition of automatically activated stereotype-congruent thoughts

Findings Summary

Across the three experiments, high- and low-prejudice participants showed equivalent knowledge of the Black cultural stereotype (Experiment 1) and experienced equally strong automatic activation of that stereotype when confronted with a race-unspecified target (Experiment 2). However, only low-prejudice participants were able to deliberately suppress stereotype-congruent thoughts and replace them with equality-oriented thoughts in a thought-listing task (Experiment 3), indicating that controlled processes can override the automatic response. Thus, the data converge on the claim that automatic stereotype activation is not moderated by prejudice level, but diverge in showing that controlled inhibition of those activations is contingent on participants' motivation to maintain a nonprejudiced identity.

Evidence-Claim Alignment

Supported Claims

- High- and low-prejudice individuals possess equivalent knowledge of the Black cultural stereotype (Experiment 1 finding of no difference in stereotype content across prejudice levels).
- Automatic activation of the Black stereotype occurs equally strongly for high- and low-prejudice participants when presented with a race-unspecified target (Experiment 2 result showing stereotype-congruent evaluations regardless of prejudice level).
- Low-prejudice participants can deliberately inhibit automatically activated stereotype-congruent thoughts and replace them with equality-oriented thoughts in a thought-listing task (Experiment 3 evidence of controlled inhibition unique to low-prejudice

group).

Unsupported or Weakly Supported Claims

- The paper's central claim that stereotypes and personal beliefs are distinct constructs with different implications for prejudice is not empirically substantiated, as none of the three experiments measured personal belief endorsement separately from stereotype knowledge.
- The assertion that controlled processes can generally override automatic prejudice in everyday intergroup settings is only demonstrated within the constrained laboratory thought-listing paradigm, leaving its broader applicability ambiguous.

Contradictions or Tensions

- The introduction foregrounds the distinction between stereotypes and personal beliefs, yet the experimental work assesses only stereotype knowledge and automatic activation, creating a mismatch between the theoretical premise and the operationalized variables.
- The authors argue that their findings refute the inevitability-of-prejudice framework, but they provide no direct comparative test pitting that framework against their model, resulting in a theoretical rebuttal that is not empirically grounded.

Strengths

- Clear empirical test of the core assumption that high- and low-prejudice individuals possess equivalent knowledge of the Black cultural stereotype (Experiment 1).
- Use of a free-response listing task to assess stereotype content without cueing, enhancing ecological validity of stereotype knowledge measurement.
- Converging evidence across three experiments that automatic stereotype activation is equally strong for high- and low-prejudice participants (Experiment 2) and that controlled processes can suppress stereotype-congruent thoughts in low-prejudice participants (Experiment 3).
- Integration of classic information-processing models (Posner & Snyder, 1975; Schneider & Shiffrin, 1977) to frame automatic vs. controlled components of prejudice, providing a theoretically grounded dual-process perspective.
- Transparent reporting of participant characteristics, stimulus materials, and coding procedures for thought-listing, supporting reproducibility of the core tasks.
- Discussion explicitly links findings to broader theoretical debates (inevitability of prejudice vs. limited-capacity processing) and outlines concrete future directions for modeling controlled inhibition.

Weaknesses and Concerns

- The central theoretical distinction between stereotypes and personal beliefs is never operationalized; none of the three experiments measure personal belief endorsement, leaving a key construct untested.
- Absence of a priori power analysis and limited sample size (N = 40 in Experiment 1) raise concerns about Type II error, especially for null findings regarding prejudice differences.
- Methodological details are sparse: randomization, counterbalancing, stimulus timing, and exclusion criteria are not reported, hindering assessment of potential order or selection biases.
- The paper does not engage with major alternative dual-process accounts (e.g., Fazio's MODE model), limiting the theoretical scope and overstating the uniqueness of its model.
- Claims that the findings refute the inevitability-of-prejudice framework lack direct comparative evidence; the rebuttal is purely theoretical rather than empirically demonstrated.
- Experiment 1's introduction fails to state explicit hypotheses, creating a minor inconsistency between the reported prediction testing and the framing of the study.

Key References

- **Posner & Snyder (1975)** -- foundational theory
- **Schneider & Shiffrin (1977)** -- foundational theory
- **Shiffrin & Schneider (1977)** -- foundational theory
- **Allport (1954)** -- competing theoretical account
- **Billig (1985)** -- competing theoretical account
- **Ehrlich (1973)** -- key empirical precedent
- **Crosby et al. (1980)** -- competing theoretical account
- **Logan (1980)** -- primary methodological precedent
- **Logan & Cowan (1984)** -- primary methodological precedent
- **Neely (1977)** -- primary methodological precedent
- **Devine (1988)** -- key empirical precedent
- **Harding, Proshansky, Kutner, & Chein (1969)** -- foundational theory
- **Secord & Backman (1974)** -- foundational theory
- **Bargh (1984)** -- primary methodological precedent

- **Fazio (1995)** -- competing theoretical account

Abstract

Preserved from source without summarization.

Three studies tested basic assumptions derived from a theoretical model based on the dissociation of automatic and controlled processes involved in prejudice. Study 1 supported the model's assumption that high- and low-prejudice persons are equally knowledgeable of the cultural stereotype. The model suggests that the stereotype is automatically activated in the presence of a member (or some symbolic equivalent) of the stereotyped group and that low-prejudice responses require controlled inhibition of the automatically activated stereotype. Study 2, which examined the effects of automatic stereotype activation on the evaluation of ambiguous stereotype-relevant behaviors performed by a race-unspecified person, suggested that when subjects' ability to consciously monitor stereotype activation is precluded, both high- and low-prejudice subjects produce stereotype-congruent evaluations of ambiguous behaviors. Study 3 examined high- and low-prejudice subjects' responses in a consciously directed thought-listing task. Consistent with the model, only low-prejudice subjects inhibited the automatically activated stereotype-congruent thoughts and replaced them with thoughts reflecting equality and negations of the stereotype. The relation between stereotypes and prejudice and implications for prejudice reduction are discussed.

Introduction (Paper-Level)

Section: paper-level

Research Gap / Motivation

The existing literature has not adequately captured the distinction between stereotypes and personal beliefs nor explored how each contributes to responses toward stereotyped groups.

Theoretical Framework

automatic vs controlled processes model

This work challenges the inevitability of prejudice framework and offers a model of responses to members of stereotyped groups that is derived largely from work in information processing that distinguishes between automatic (mostly involuntary) and controlled (mostly voluntary) processes (e.g., Posner & Snyder, 1975; Schneider & Shiffrin, 1977; Shiffrin & Schneider, 1977).

- Citations: Posner & Snyder (1975); Schneider & Shiffrin (1977); Shiffrin & Schneider (1977)

distinction between stereotypes and personal beliefs

Although they may have some overlapping features, it is argued that stereotypes and personal beliefs are conceptually distinct cognitive structures. Each structure represents part of one's entire knowledge base of a particular group.

Mechanistic Claims

different implications of stereotypes vs beliefs

To the extent that stereotypes and personal beliefs represent different and only potentially overlapping subsets of information about ethnic or racial groups, they may have different implications for evaluation of and behavior toward members of the ethnic and racial groups.

Constructs of Interest

- **stereotypes** (*predictor*): Cognitive structures representing knowledge about a group that may be automatically activated.
- **personal beliefs** (*predictor*): Propositions endorsed as true about a group, potentially distinct from stereotype knowledge.
- **responses toward stereotyped groups** (*outcome*): Evaluations and behaviors directed at members of groups for which stereotypes exist.

Rationale for Predictor

Stereotypes have been theorized to drive prejudice automatically, while personal beliefs may moderate or override that influence, making them critical to understand the origins of discriminatory responses.

Rationale for Outcome

Understanding how people respond to stereotyped groups is essential for elucidating the mechanisms linking cognition (stereotypes, beliefs) to prejudice and behavior.

Key Terminology

automatic processes: Unintentional or spontaneous activation of well-learned associations or responses that occur without conscious effort and are triggered by stimulus cues.

Automatic processes involve the unintentional or spontaneous activation of some well-learned set of associations or responses that have been developed through repeated activation in memory.

controlled processes: Intentional, effortful processes that require active attention and are flexible, supporting decision making and problem solving.

Controlled processes are intentional and require the active attention of the individual. Controlled processes, although limited by capacity, are more flexible than automatic processes.

Background Research

- Harding et al. (1969) and Secord & Backman (1974) described stereotypes as the cognitive component of prejudiced attitudes.
- Allport (1954), LaViolette & Silvert (1951), Saenger (1953), and Simpson & Yinger (1965) suggested stereotypes function to rationalize prejudice.
- Allport (1954), Billig (1985), Ehrlich (1973), Hamilton (1981), and Tajfel (1981) argued that prejudice is an inevitable consequence of ordinary categorization processes.
- Ashmore & Del Boca (1981) and Billig (1985) highlighted the distinction between knowledge of a stereotype and endorsement of it.
- Bettelheim & Janowitz (1964), Brigham (1972), Devine (1988), and Karlins et al. (1969) found no significant relation between reported stereotypes and prejudice in interview studies.
- Posner & Snyder (1975), Schneider & Shiffrin (1977), and Shiffrin & Schneider (1977) outlined automatic and controlled information-processing processes.
- Logan (1980), Logan & Cowan (1984), Neely (1977), and Posner & Snyder (1975) provided evidence that automatic and controlled processes can operate independently.
- Neely (1977) demonstrated how conscious expectancies can inhibit automatically activated responses in a lexical decision task.

Alternative Theories Discussed

inevitability of prejudice framework

- **Description:** The view that as long as stereotypes exist, prejudice will automatically follow; knowledge of a stereotype is equated with prejudice toward the group.
- **Difference from focal theory:** Unlike the automatic-controlled model that separates involuntary activation from voluntary regulation and distinguishes stereotypes from personal beliefs, the inevitability framework treats stereotype knowledge and prejudice as synonymous and does not account for personal endorsement.
- **Citations:** Allport (1954); Billig (1985); Ehrlich (1973)

Overview of Studies

The paper previews three studies that examine how stereotypes and personal beliefs are involved in responses toward stereotyped groups.

Missing or Unclear

- Specific, testable hypotheses derived from the proposed model
- Explicit predictions linking particular measures of stereotypes or beliefs to response outcomes

All Citations Referenced

- **Harding, Proshansky, Kutner, & Chein (1969)** -- tripartite model of attitudes
- **Secord & Backman (1974)** -- tripartite model of attitudes
- **Allport (1954)** -- stereotypes functional for prejudice
- **LaViolette & Silvert (1951)** -- stereotypes functional for prejudice
- **Saenger (1953)** -- stereotypes functional for prejudice
- **Simpson & Yinger (1965)** -- stereotypes functional for prejudice
- **Billig (1985)** -- inevitability of prejudice
- **Ehrlich (1973)** -- inevitability of prejudice
- **Hamilton (1981)** -- inevitability of prejudice
- **Tajfel (1981)** -- inevitability of prejudice
- **Ashmore & Del Boca (1981)** -- distinction between knowledge and endorsement
- **Bettleheim & Janowitz (1964)** -- no relation between stereotypes and prejudice
- **Brigham (1972)** -- no relation between stereotypes and prejudice
- **Devine (1988)** -- no relation between stereotypes and prejudice

- **Karlins, Coffman, & Walters (1969)** -- no relation between stereotypes and prejudice
- **Posner & Snyder (1975)** -- automatic vs controlled processes
- **Schneider & Shiffrin (1977)** -- automatic vs controlled processes
- **Shiffrin & Schneider (1977)** -- automatic vs controlled processes
- **Logan (1980)** -- independent automatic and controlled processes
- **Logan & Cowan (1984)** -- independent automatic and controlled processes
- **Neely (1977)** -- automatic vs controlled processes
- **Shiffrin & Dumais (1981)** -- automatic processes inattention
- **Neely (1977)** -- semantic priming and expectancy inhibition

Additional Quoted Evidence

This work challenges the inevitability of prejudice framework and offers a model of responses to members of stereotyped groups that is derived largely from work in information processing that distinguishes between automatic (mostly involuntary) and controlled (mostly voluntary) processes.

The basic argument of the inevitability of prejudice perspective is that as long as stereotypes exist, prejudice will follow.

Experiment 1 -- Introduction

Section: exp_1

Rationale for This Experiment

The experiment aims to validate the present model by directly examining whether high- and low-prejudice subjects differ in their knowledge of the cultural stereotype of Blacks.

Relation to Prior Experiments

This is the first experiment in the series, establishing baseline knowledge of stereotype content before later experiments manipulate other variables.

Theoretical Refinements (relative to paper-level intro)

individual differences in prejudice knowledge

Historically, little attention has focused on individual differences in prejudice when assessing the content of stereotypes.

New Elements Introduced

- Use of a free response task instead of the typical adjective checklist assessment
- Assessment of stereotype content without providing cue lists

Constraints Addressed

- Addresses the assumption that high- and low-prejudice individuals are equally knowledgeable about cultural stereotypes

Missing or Unclear

- Specific predictions tested in this experiment are not explicitly stated in the introduction.

Citations Referenced

- **Brigham (1971)** -- implicit in the stereotype assessment literature
- **Gilbert (1951)** -- typical adjective checklist assessment of stereotype content
- **Karlins et al. (1969)** -- typical adjective checklist assessment of stereotype content
- **Katz & Braly (1933)** -- typical adjective checklist assessment of stereotype content

Experiment 1 -- Methods

Section: exp_1

Participants

- **N:** 40
- **N per condition:** 21 (high-prejudice), 19 (low-prejudice)
- **Demographics:** White introductory psychology students
- **Recruitment:** introductory psychology student pool
- **Compensation:** course credit
- **Testing context:** tested in groups of 4-6 in a lab; subjects isolated from each other and experimenter left the room after instructions

Design

Design statement: 2 (prejudice level: high vs low) between-subjects design.

Predictors:

- **prejudice level**
- Levels: high, low
- Manipulation: between-subjects
- Factor type: nominal

Counterbalancing: not reported

Outcome Measures

- **proportion of thoughts listed in each coding category:** percentage of listed stereotype components falling into each predefined coding category, calculated separately for high- and low-prejudice groups

Materials

- **Description:** Paper questionnaire with blank lines for participants to list components of the Black cultural stereotype; 7-item Modern Racism Scale
- **Source:** Modern Racism Scale (McConahay, Hardee, & Batts, 1981); other items created by authors
- **Additional details:**
 - Modern Racism Scale uses a 5-point rating from -2 (disagree strongly) to +2 (agree strongly)
 - Scale range -14 to +14; Cronbach's alpha = .83

Procedure

Ordered steps:

1. Experimenter gave general written instructions about the questionnaire purpose
 2. Participants listed components of the Black cultural stereotype on a blank page
 3. Participants completed the seven-item Modern Racism Scale
 4. Completed booklet placed in an unmarked envelope and deposited in a box
 5. Debriefing and thanks
- **Apparatus:** paper-and-pencil questionnaires in a laboratory setting

- **Experimenter presence:** experimenter left the room after giving general instructions
- **Presentation format:** not reported
- **Stimulus timing:** not reported
- **Session duration:** not reported

Verbatim instructions (as quoted in the paper):

Written instructions told subjects that the questionnaire was designed to help researchers better understand social stereotypes and that interest centered on the cultural stereotype of Blacks.

The experimenter informed them that she was not interested in their personal beliefs but in their knowledge of the content of the cultural stereotype.

Preregistration and Analysis Plan

- **Preregistration:** not reported

Missing or Unclear

- no power analysis reported
- no randomization or counterbalancing details for stimulus presentation
- no detailed stimulus timing information
- no explicit exclusion criteria
- limited demographic details (age, gender, education not reported)
- no description of apparatus beyond paper questionnaires
- no session length reported
- no preregistration information

Citations Referenced in This Section

- **McConahay, Hardee, & Batts (1981)** -- source of the Modern Racism Scale used as a measure of anti-Black attitudes
- **Kinder & Sears (1981)** -- supporting claim that the Modern Racism Scale predicts voting patterns and reactions to busing
- **Sears & Kinder (1971)** -- supporting claim that the Modern Racism Scale predicts voting patterns and reactions to busing

- **Sears & McConahay (1973)** -- supporting claim that the Modern Racism Scale predicts voting patterns and reactions to busing

Quoted Evidence from Source

Forty White introductory psychology students participated in groups of 4-6 for course credit.

Subjects were assigned to a high-prejudice ($N = 21$) or a low-prejudice ($N = 19$) group on the basis of a median split of scores on the scale.

Written instructions told subjects that the questionnaire was designed to help researchers better understand social stereotypes and that interest centered on the cultural stereotype of Blacks.

Experiment 1 -- Results and Discussion

Section: exp_1

Statistical Approach

The prediction was tested using chi-square tests on the differences in Table 1 and judges' predictions of prejudice level from the content of the protocols.

Key Findings

High- and low-prejudice subjects did not differ in their knowledge of the cultural stereotype.

- **Test:** Chi-square tests and judges' predictions
- **Effect type:** main effect
- **Variables in model:** prejudice level, stereotype content

Verbatim statistical report:

none of the differences in Table 1 was statistically reliable. The judges could not reliably predict the subjects' prejudice level from the content of their protocols.

- Citations: Ehrlich (1973)

Authors' Interpretations

Aggressiveness in stereotypes

The most striking aspect of these data is that the most common theme in subjects' protocols was that Blacks are aggressive, hostile, or criminal-like (see Table 1). All subjects listed either the aggressive or criminal categories and many listed both categories. This finding is important because, as was suggested earlier, much of the intergroup perception literature has been predicated on the assumption that Blacks are hostile and aggressive.

- Citations: Donnerstein & Donnerstein (1972); Donnerstein, Donnerstein, Simon, & Ditrachs (1972); Duncan (1976); Sager & Schofield (1980)

Stereotype content

Consistent with the stereotype assessment literature, the protocols were dominated by trait listings and were predominately negative.

Prejudice and stereotype knowledge

These data validate Ehrlich's (1973) assumption as well as the first assumption of the present model: High- and low-prejudice persons are indeed equally knowledgeable of the cultural stereotype.

- Citations: Ehrlich (1973)

Authors' Stated Links to Their Own Predictions

No difference in stereotype knowledge

The prediction of no difference between the high- and low-prejudice subjects' knowledge of the cultural stereotype was tested in two different ways.

Missing or Unclear (Non-Statistical)

- Details about the specific chi-square tests and judges' prediction accuracy rates are not provided.

Citations Referenced

- **Donnerstein & Donnerstein (1972)** -- research on the effect of racial stereotypes on perception and behavior
- **Donnerstein, Donnerstein, Simon, & Ditrachs (1972)** -- research on the effect of racial stereotypes on perception and behavior

- **Duncan (1976)** -- research on the effect of racial stereotypes on perception and behavior
- **Sager & Schofield (1980)** -- research on the effect of racial stereotypes on perception and behavior
- **Ehrlich (1973)** -- assumption about stereotype knowledge

General Discussion

Section: paper-level

Summary of Findings

The discussion reports that both high- and low-prejudice individuals possess stereotype knowledge and show equally strong automatic activation, but controlled processes can inhibit prejudice-like responses when goals to be nonprejudiced are salient; change requires intentional, effortful control, and prejudice is not inevitable but a product of limited processing capacity.

Theoretical Conclusions

controlled processes inhibit automatic prejudice

Study 3, however, provided evidence that controlled processes can inhibit the effects of automatic processing when the implications of such processing compete with goals to establish or maintain a nonprejudiced identity.

automatic activation functions like a bad habit

Inhibiting stereotype-congruent or prejudice-like responses and intentionally replacing them with nonprejudiced responses can be likened to the breaking of a bad habit.

change requires intention, attention, and time

In summary, at minimum, the attitude and belief change process requires intention, attention, and time.

prejudice not inevitable, limited-capacity processing

Thus, in contrast to the pessimistic analysis by Crosby et al. (1980), the present framework suggests that rather than all people being prejudiced, all are victims of being limited capacity processors.

- Citations: Crosby et al. (1980)

Cross-Experiment Integration

automatic activation equal across prejudice levels

Study 2 suggested that automatic stereotype activation is equally strong and equally inescapable for high- and low-prejudice subjects.

controlled processes can override automatic effects

Study 3, however, provided evidence that controlled processes can inhibit the effects of automatic processing when the implications of such processing compete with goals to establish or maintain a nonprejudiced identity.

Results Authors State Support Their Theory

knowledge of stereotype present in both prejudice groups

The model examined in these studies makes a clear distinction between knowledge of the racial stereotype, which Study 1 suggested both high- and low-prejudice persons possess, and personal beliefs about the stereotyped group.

controlled processes can inhibit prejudice-like responses

Study 3, however, provided evidence that controlled processes can inhibit the effects of automatic processing when the implications of such processing compete with goals to establish or maintain a nonprejudiced identity.

Results Authors State Challenge a Theory

challenge to inevitability of prejudice premise

In the context of the present model in which automatic processes and controlled processes can be dissociated, I disagree fundamentally with this premise. Such an argument denies the possibility for change in one's attitudes and beliefs, and I view this as a severe limitation of the Crosby et al. (1980) analysis.

- Citations: Crosby et al. (1980)

Alternative Accounts Addressed by Authors

Crosby et al. (1980) inevitability analysis (*named theory*)

- **Description:** Nonconsciously monitored responses are more trustworthy than consciously mediated responses; nonprejudiced behavior is impression management rather

than genuine attitude change.

- **How authors addressed it:** argued against; claim it denies possibility of attitude change and is a limitation
- **Citations:** Crosby et al. (1980)

Implications for the Field

need for precise prejudice labeling criteria

What this analysis requires is that theoreticians be more precise on the criteria established for labeling behavior as prejudiced or nonprejudiced.

explore variables that engage controlled inhibition

The present framework highlights the potential for nonprejudiced behaviors when social desirability concerns are minimal (Study 3) and invites researchers to explore the variables that are likely to engage controlled stereotype-inhibiting processes in intergroup settings.

Limitations Stated by Authors

- The present model and set of empirical studies certainly does not resolve the issue of precise criteria for labeling behavior as prejudiced or nonprejudiced.
- A fully articulated model of controlled processes that delineates the cognitive mechanisms involved in inhibition is still needed.

Future Directions Proposed by Authors

- Develop a detailed model of controlled processes that explains the inhibition mechanism (e.g., drawing on Logan and Cowan, 1984).
- Systematically explore variables that engage controlled stereotype-inhibiting processes in intergroup settings.
- Investigate how intention, attention, and time constraints affect the success of prejudice reduction.

Citations Referenced

- **Pettigrew (1987)** --
- **Crosby et al. (1980)** --

- Baxter (1973) --
- Gaertner (1976) --
- Gaertner & Dovidio (1977) --
- Linn (1965) --
- Weitz (1972) --
- Logan and Cowan (1984) --
- Bargh (1984) --
- Ronis, Yates, and Kirscht (in press) --
- Greeley & Sheatsley (1971) --
- Taylor et al. (1978) --
- Higgins & King (1981) --

Critique

Paper: devine-1989-stereotypes-and-prejudice--their-automatic-and-controlled-components

Overall Assessment

The paper demonstrates moderate methodological rigor but is weakened by several notable gaps, including the absence of a power analysis, insufficient description of randomization and counterbalancing, and a lack of clarity about exclusion criteria. Its theoretical framework is under-specified, failing to integrate well-established dual-process models and offering limited justification for dismissing alternative accounts of prejudice regulation. Consequently, the evidentiary support for the central claims is tentative, as the experiments do not directly assess the personal-belief construct highlighted in the introduction. Overall, these high-severity concerns limit the paper's standing as a robust contribution to the literature on automatic and controlled components of prejudice.

3. Internal Consistency

- **The introduction emphasizes the distinction between stereotypes and personal beliefs, yet none of the three experiments directly assess personal beliefs as a separate construct.**
- Severity: **moderate**
- Justification: The paper's theoretical framework posits that stereotypes and personal beliefs may have different implications for prejudice. However, the empirical work presented

(Experiments 1-3) only measures stereotype knowledge, automatic activation, and controlled inhibition, leaving personal beliefs unexamined. This gap creates a mismatch between the stated constructs of interest and the actual variables operationalized, potentially limiting the support for the broader theoretical claim.

Evidence:

- introduction -- field constructs_of_interest: Defines "personal beliefs" as a predictor distinct from stereotypes
- exp_introduction/exp_1 -- field rationale_for_this_experiment: Focuses on knowledge of the cultural stereotype, not personal beliefs
- exp_introduction/exp_2 -- field rationale_for_this_experiment: Examines automatic stereotype activation on ambiguous behavior evaluation, without measuring personal belief endorsement
- exp_introduction/exp_3 -- field rationale_for_this_experiment: Investigates controlled inhibition of automatically activated thoughts, but does not assess participants' personal belief structures

Experiment 1's introduction does not articulate explicit predictions, yet the results section reports that a specific prediction (no difference in stereotype knowledge) was tested and confirmed.

- Severity: **minor**
- Justification: The lack of clearly stated hypotheses in the Experiment 1 introduction makes it unclear whether the reported test was pre-specified. While the results claim to have tested a prediction, the introductory material does not present that prediction, leading to a minor inconsistency between the experiment's framing and its reported analyses.
- Evidence:
 - exp_introduction/exp_1 -- field missing_or_unclear: Specific predictions tested in this experiment are not explicitly stated in the introduction.
 - results_and_discussion/exp_1 -- field authors_stated_link_to_predictions: The prediction of no difference between high- and low-prejudice subjects' knowledge of the cultural stereotype was tested...

4. Alternative Accounts

- The authors do not consider well-established dual-process attitude models (e.g., Fazio's MODE model) that also explain automatic activation and controlled regulation of prejudice.

- Severity: **moderate**
- Justification: Fazio's MODE model and related dual-process accounts provide a competing explanation for how automatic stereotype activation can be overridden by motivation and opportunity. By not acknowledging or testing these alternatives, the paper limits the theoretical scope of its conclusions and may overstate the uniqueness of its proposed model.

Evidence:

- introduction -- field alternative_theories: Only the "inevitability of prejudice" framework is listed as an alternative; other major dual-process accounts are absent.
- general_discussion -- field alternative_accounts_addressed: The discussion only addresses Crosby et al. (1980) and does not evaluate competing dual-process theories.

The inevitability of prejudice framework is dismissed without presenting direct empirical evidence that controlled processes can override automatic prejudice in the contexts described.

- Severity: **moderate**
- Justification: While the authors claim that controlled inhibition disproves the inevitability view, they do not present a direct test that pits the inevitability account against their model. This weakens the claim that their findings uniquely refute the alternative.
- Evidence:
- general_discussion -- field alternative_accounts_addressed: Authors argue against Crosby et al. (1980) but provide only a theoretical rebuttal, not data directly contrasting the two accounts.

6. Methodological Concerns

- **No power analysis was reported despite a modest total sample size of 40 participants**
- Severity: **moderate**
- Justification: The study relies on a between-subjects comparison with 21 high-prejudice and 19 low-prejudice participants, but without a priori power calculations it is unclear whether the sample is sufficient to detect the expected effects. This limits confidence in null findings and raises concerns about Type II error.

Evidence:

- methods/exp_1 -- field power_analysis: power_analysis field is null and missing_or_unclear notes 'no power analysis reported'

Counterbalancing and randomization procedures for stimulus presentation are not described

- Severity: **moderate**
- Justification: For experiments that involve free-response tasks or any ordering of items, lack of information about how items were randomized or counterbalanced makes it difficult to assess potential order effects or biases in the data collection.

Evidence:

- methods/exp_1 -- field design: counterbalancing field states 'not reported'
- methods/exp_1 -- field procedure: missing details on stimulus timing and presentation format

Exclusion criteria are not specified, leaving participant selection ambiguous

- Severity: **minor**
- Justification: Without clear exclusion rules (e.g., for incomplete data or extreme scores), readers cannot evaluate whether the final sample was appropriately screened, which may affect the internal validity of the findings.

Evidence:

- methods/exp_1 -- field participants: exclusions array is empty and missing_or_unclear notes 'no explicit exclusion criteria'

The study does not mention preregistration or open data practices

- Severity: **minor**
- Justification: Absence of a preregistration statement limits transparency about the confirmatory versus exploratory nature of the analyses, which is increasingly expected for rigorous empirical work.
- Evidence:
- methods/exp_1 -- field preregistration_status: preregistration_status is 'not reported'